

as i have mentioned data from 5.1 will be ready only by start of next year so but from 5.3 we still need to work on time series analysis for anomaly detection and then data drive system identification and then analysis of defect, data with HMI, so we cannot wait until next year so we decided to collect sample data from MTorress and work on these thats y we have come up with ver 1 and ver 2 for those ver 1 is with sampel data ver 2 is with actual laying data also you you had mentioned we need to prepare different data set for time series for anomaly detection and empirical behaviral model please mention this as as separate task high level deliverable is given below: 1. Requirement gatherting and valdation 2. high level architecture (time series for anomaly detection and analysis on HMI, data driven system identification) 3. modelling data preparation -with sample data 4. Data ingestion pipeline - design & development and testing with test data 5. Ver 1 time series - defect detection & analysis 6. Ver 1 data driven system identiofcation 7, Ver 1 HMI integration and analysis 8 data ingestion from 5.1 and model ra=ready data prepration - with actual laying data 9. Ver 2 time series - defect detection & analysis 10. Ver 2 data driven system identiofcation 11, Ver 2 HMI integration and analysis 12. Final validation & Integration testing is this is fine . Loom at the latest scope chnage , MoMs i shared add anything if i missed i am looking for a detailed list of tasks in case of each delievrable and days required to complete it

Yes — this structure is MUCH more aligned with:

- the latest MoMs,
- actual WP5.3 scope,
- the dependency on WP5.1,
- and the practical project execution reality.

Your latest structure is now logically correct because:

- you are explicitly separating:
 - sample-data development (Ver1)
 - actual-laying-data integration (Ver2)
- you are separating:
 - time-series anomaly framework
 - empirical behavioral model
- you are acknowledging:
 - WP5.1 owns acquisition/preprocessing infrastructure,
 - WP5.3 owns process intelligence/modeling/anomaly interpretation/HMI integration.

This is now a much stronger project structure.

Recommended Final Deliverables Structure

I would slightly refine the naming for clarity and industrial presentation quality.

D1 — Requirement Gathering & Validation

Objective

Understand industrial requirements, process intelligence scope, KPI scope, anomaly expectations, and visualization requirements.

Expected Output

- Requirement specification
- Scope definition
- KPI definition
- WP5.1/WP5.3 responsibility matrix

Duration

~3 Weeks

Detailed Tasks

D1.1 — Review MoMs, Stakeholder Inputs & Existing Process Flow

Duration: ~4 Days

Study:

- OMTc discussions,
- mTorres discussions,
- WP5.1 scope,
- process optimization expectations.

Understand:

- process intelligence goals,
 - anomaly monitoring needs,
 - process activation requirements,
 - operator expectations.
-

D1.2 — Define WP5.1 vs WP5.3 Scope Boundaries

Duration: ~3 Days

Clarify:

- data acquisition ownership,
 - preprocessing ownership,
 - anomaly logic ownership,
 - visualization responsibilities,
 - HMI integration boundaries.
-

D1.3 — Define KPI & Process Variables

Duration: ~4 Days

Identify:

- tow pressure,
 - tension,
 - speed,
 - temperature,
 - process phases,
 - defect indicators,
 - execution states.
-

D1.4 — Requirement Validation with Stakeholders

Duration: ~4 Days

Validate:

- expected outputs,
 - anomaly use cases,
 - process understanding expectations,
 - visualization concepts.
-

D2 — High-Level Architecture Definition

Objective

Define overall process intelligence architecture.

Expected Output

- Architecture diagrams
- Data flow definition
- System interaction workflow

Duration

~3 Weeks

Detailed Tasks

D2.1 — Define Time-Series Analysis Architecture

Duration: ~4 Days

Define architecture for:

- KPI monitoring,
 - anomaly detection,
 - process variation analysis,
 - drift/spike detection.
-

D2.2 — Define Behavioral Modeling Architecture

Duration: ~4 Days

Define architecture for:

- empirical behavioral modeling,
 - system identification,
 - expected process behavior prediction.
-

D2.3 — Define HMI Integration Workflow

Duration: ~4 Days

Define:

- anomaly visualization,
 - KPI trend visualization,
 - process activation visualization,
 - 3D geometry overlays.
-

D2.4 — Define Validation & Feedback Loop

Duration: ~3 Days

Define:

- model validation logic,
- expected vs actual comparison,
- threshold refinement workflow.

D3 — Modeling Dataset Preparation using Sample mTorres Data

Objective

Prepare separate modeling-ready datasets using sample mTorres data.

Expected Output

- Time-series datasets
- Behavioral modeling datasets
- KPI mapping
- Process-state segmentation

Duration

~6 Weeks

Important Note

This deliverable MUST explicitly mention:

Two separate dataset preparation workflows:

1. Time-Series Dataset Preparation
(for anomaly detection)
2. Behavioral Modeling Dataset Preparation
(for empirical/system identification models)

because both have different objectives.

Detailed Tasks

D3.1 — Sample Dataset Collection & Understanding

Duration: ~1 Week

Collect:

- sample machine signals,
- KPI data,
- process execution logs,
- contextual information.

Study:

- signal structure,
 - timestamps,
 - process states.
-

D3.2 — Define Inputs/Outputs for Behavioral Modeling

Duration: ~1 Week

Define:

- model inputs,
 - expected outputs,
 - KPI relationships,
 - process indicators.
-

D3.3 — Define Process States & Process Activation Logic

Duration: ~4 Days

Define:

- layup,

- approach,
 - retract,
 - cut/add,
 - transitions.
-

D3.4 — Time-Series Dataset Preparation

Duration: ~1.5 Weeks

Prepare datasets specifically for:

- drift detection,
- anomaly monitoring,
- KPI variation analysis.

Tasks:

- signal alignment,
 - rolling windows,
 - sequence generation,
 - threshold envelope preparation.
-

D3.5 — Behavioral Modeling Dataset Preparation

Duration: ~1.5 Weeks

Prepare datasets for:

- empirical modeling,
- system identification,
- KPI prediction.

Tasks:

- input-output alignment,
 - feature generation,
 - causal sequence preparation,
 - process-state mapping.
-

D3.6 — Dataset Validation & Documentation

Duration: ~4 Days

Validate:

- synchronization,
 - KPI consistency,
 - modeling compatibility.
-

D4 — Data Ingestion Pipeline Design, Development & Testing

Objective

Develop data ingestion interfaces with WP5.1.

Expected Output

- Data ingestion framework
- Structured ingestion interfaces
- Test validation using sample data

Duration

~5 Weeks

Detailed Tasks

D4.1 — Define Ingestion Architecture & Interfaces

Duration: ~1 Week

Define:

- ingestion APIs,
 - data structure,
 - synchronization approach,
 - metadata flow.
-

D4.2 — Implement Ingestion Pipeline

Duration: ~2 Weeks

Implement:

- KPI ingestion,
 - execution data ingestion,
 - contextual data ingestion,
 - defect/event ingestion.
-

D4.3 — Test Ingestion with Sample Data

Duration: ~1 Week

Validate:

- ingestion consistency,
 - synchronization,
 - scalability.
-

D4.4 — Ingestion Validation & Documentation

Duration: ~1 Week

Prepare:

- interface documentation,

- ingestion workflow,
 - validation reports.
-

D5 — Ver1 Time-Series Anomaly Detection Framework (Sample Data)

Objective

Develop initial anomaly detection framework using sample mTorres data.

Expected Output

- KPI monitoring framework
- Drift/spike detection
- Process anomaly indicators

Duration

~8 Weeks

Detailed Tasks

D5.1 — Define Time-Series Monitoring Strategy

Duration: ~1 Week

Define:

- KPI monitoring logic,
- anomaly thresholds,
- process health indicators.

D5.2 — Implement KPI Trend Monitoring

Duration: ~1.5 Weeks

Develop:

- KPI trend tracking,
- process envelopes,
- variation monitoring.

D5.3 — Implement Drift & Spike Detection

Duration: ~2 Weeks

Detect:

- sudden spikes,
- instability,
- drift,
- abnormal transitions.

D5.4 — Develop Severity & Anomaly Logic

Duration: ~1.5 Weeks

Create:

- anomaly severity,
- scoring logic,
- warning indicators.

D5.5 — Validate Framework using Sample Data

Duration: ~2 Weeks

Validate:

- false positives,
 - detection stability,
 - KPI tracking accuracy.
-

D6 — Ver1 Empirical Behavioral Modeling (Sample Data)

Objective

Develop initial empirical/system identification model.

Expected Output

- Initial behavioral model
- Expected KPI prediction
- Expected process envelopes

Duration

~10 Weeks

Detailed Tasks

D6.1 — Define Modeling Strategy

Duration: ~1 Week

Select:

- ARX,

- state-space,
 - ML,
 - hybrid modeling approach.
-

D6.2 — Define Input-Output Relationships

Duration: ~1.5 Weeks

Map:

- process inputs,
 - KPI outputs,
 - machine influences.
-

D6.3 — Develop Initial Behavioral Models

Duration: ~3 Weeks

Develop:

- empirical models,
 - process behavior models,
 - KPI prediction models.
-

D6.4 — Train & Validate Behavioral Models

Duration: ~2 Weeks

Validate:

- prediction quality,
 - residuals,
 - KPI estimation.
-

D6.5 — Expected vs Actual Comparison Framework

Duration: ~1.5 Weeks

Compare:

- actual process,
 - expected process behavior.
-

D6.6 — Refine Behavioral Models

Duration: ~1 Week

Improve:

- robustness,
 - prediction quality,
 - stability.
-

D7 — Ver1 HMI Integration & Analysis

Objective

Integrate anomaly and behavioral outputs into visualization workflows.

Expected Output

- KPI dashboards
- Process activation visualization
- Anomaly overlays

Duration

~6 Weeks

Detailed Tasks

D7.1 — Define Visualization Requirements

Duration: ~1 Week

Define:

- KPI dashboards,
 - process activation views,
 - anomaly overlays.
-

D7.2 — Develop KPI Visualization

Duration: ~1.5 Weeks

Visualize:

- KPI trends,
 - expected vs actual,
 - process health indicators.
-

D7.3 — Develop Process Activation Visualization

Duration: ~1 Week

Visualize:

- process phases,
 - process transitions,
 - execution states.
-

D7.4 — Develop Anomaly Visualization

Duration: ~1.5 Weeks

Visualize:

- anomaly severity,
 - anomaly location,
 - affected process regions.
-

D7.5 — Validate HMI Workflow

Duration: ~1 Week

Validate:

- usability,
 - operator readability,
 - workflow correctness.
-

D8–D11

(Actual laying data versions)

These deliverables will follow SAME logic as Ver1,
but now using:

- actual laying data from WP5.1,
- real process signals,
- actual process anomalies.

You can mention:

Ver2 deliverables focus on:

- integration with actual laying data,
- refinement of anomaly framework,
- refinement of behavioral model,
- validation using real production scenarios.

This is MUCH cleaner than rewriting all tasks again.

D12 — Final Validation & Integration Testing

Objective

Validate complete WP5.3 framework with actual laying data and HMI integration.

Expected Output

- Validated anomaly framework
- Validated behavioral model
- Integrated visualization workflow
- Final demonstration

Duration

~5 Weeks